

# False confidence, imprecise probability, and valid statistical inference

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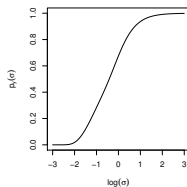
<sup>2</sup>[www4.stat.ncsu.edu/~rgmarti3](http://www4.stat.ncsu.edu/~rgmarti3)

- Common setup:
  - Observable data  $Y$
  - Statistical model for  $Y$ :  $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$
  - No prior info about  $\theta$ , ignorant
- Goal is *probabilistic* uncertainty quantification about  $\theta$  based on observed data  $Y = y$
- (Precise) data-dependent probability distributions for  $\theta$ :
  - (default-prior) Bayesian
  - fiducial
  - ...
- *Foundations matter!* Are there risks resulting from the huge jump,  $\{\text{no prior}\} \longrightarrow \{\text{precise posterior}\}$ ?

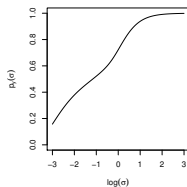
- *False confidence*: Certain false assertions about  $\theta$  will tend to be assigned high posterior prob  $\implies$  risk!
- To avoid this risk, imprecise prob considerations are needed
- Huh??
- This talk:
  - example & false confidence
  - inferential models, imprecision, and validity
  - valid IM in the example
  - concluding remarks

# Satellite collision

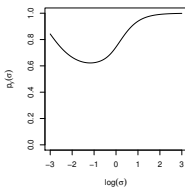
- Satellite conjunction analysis:
  - Orbiting satellite could collide with another object.
  - To avoid this, analysts compute a *non-collision probability*.<sup>3</sup>
  - Satellite judged to be safe if non-collision probability is high.
- *Noisier data makes non-collision probability large*<sup>4</sup>



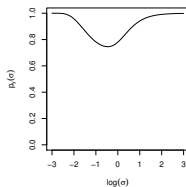
(a) Close



(b) Kinda close



(c) Kinda far



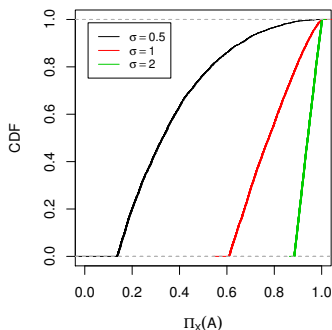
(d) Far

<sup>3</sup>Details in Balch, M., and Ferson (2019), [arXiv:1706.08565](https://arxiv.org/abs/1706.08565).

<sup>4</sup>M. (2019), [researchers.one/articles/19.02.00001](https://researchers.one/articles/19.02.00001)

# Satellite collision, cont.

- Simulated data: satellite on a collision course
- Assertion:  $A = \{\text{non-collision}\}$
- Plot shows the CDF  $\alpha \mapsto P\{\Pi_Y(A) \leq \alpha\}$ .
- Observe: non-collision probability tends to be large even though satellite's on a collision course



- *False confidence*: Hypotheses tending to be assigned high probability even if data don't support them.
- All probabilities suffer from false confidence!

False confidence theorem (Balch, M., and Ferson 2019).

Let  $\Pi_Y$  be a probability on  $\Theta$ , depending on data  $Y$ . Then, for any  $\alpha \in (0, 1)$  and any  $\theta \in \Theta$ , there exists  $A \subset \Theta$  such that

$$A \not\ni \theta \quad \text{and} \quad P_\theta\{\Pi_Y(A) > 1 - \alpha\} > \alpha.$$

- Why does this matter?
  - advantage of probabilities for inference: marginalization
  - can't reliably marginalize *everywhere*, hence risk
- Precision and reliability are too strong to be compatible
  - probabilities are too precise
  - data may not be informative enough to reliably support that level of precision without a boost from a genuine prior
- What kind of imprecision do we need?

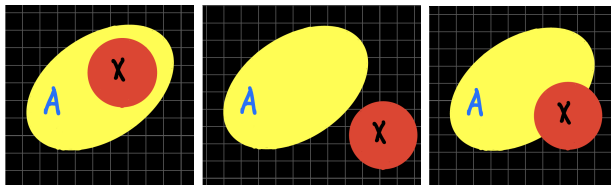
## Definition.

An *inferential model* (IM) is a function whose input is  $(y, \mathcal{P}, \dots)$  and whose output is a super-additive capacity  $\underline{\Pi}_y$ , with

$$\underline{\Pi}_y(A) = \text{degree of belief in } "\theta \in A," \text{ given } y, \quad A \subseteq \Theta$$

- “...” allows for extras, like (partial) prior information
- Includes precise/additive cases, e.g., Bayes & fiducial
- Also includes Dempster–Shafer and other genuinely imprecise/non-additive cases...





(e)  $\mathcal{X} \subseteq A$

(f)  $\mathcal{X} \subseteq A^c$

(g) otherwise

- Pics show non-additivity induced by a *random set*  $\mathcal{X}$ 
  - if  $\underline{\Pi}(A) := P(\mathcal{X} \subseteq A)$ ,
  - then  $\underline{\Pi}(A) + \underline{\Pi}(A^c) \leq 1 \leftarrow$  super-additive!
- Define the *dual* to  $\underline{\Pi}$  as  $\overline{\Pi}(A) = 1 - \underline{\Pi}(A^c)$ 
  - Super-additivity implies  $\underline{\Pi}(A) \leq \overline{\Pi}(A)$
  - $\overline{\Pi}(A) = P(\mathcal{X} \cap A \neq \emptyset)$  measures the plausibility of  $A$ .

- IM output is a pair  $(\underline{\Pi}_y, \overline{\Pi}_y)$ , lower/upper probs.
- Just like probabilities, these are personal.
- Imprecise prob literature focuses on coherence:
  - constraints on  $A \mapsto \underline{\Pi}_y(A)$  protect me from sure loss
  - internal sanity check
- More is needed to make my beliefs externally/real-world relevant, e.g., to avoid false confidence.
- “More” = constraints on  $y \mapsto \underline{\Pi}_y(A)$

## Definition.

An IM is *valid* if

$$\sup_{\theta \notin A} P_{\theta} \{ \underline{\Pi}_Y(A) > 1 - \alpha \} \leq \alpha, \quad \text{for all } A \subseteq \Theta, \alpha \in [0, 1]$$

- Validity controls the frequency at which the IM assigns relatively high beliefs to false assertions.
- Protects us from false confidence!
- “For all  $A$ ” is important for marginalization, etc.

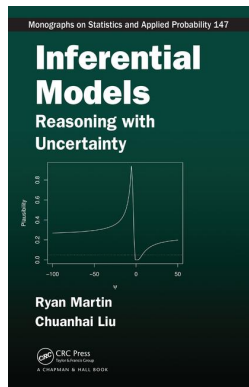
## Theorem.

If the IM is valid, then derived procedures control error rates:

- “reject  $H_0 : \theta \in A$  if  $\overline{\Pi}_y(A) \leq \alpha$ ” is a size  $\alpha$  test,
  - the  $100(1 - \alpha)\%$  plausibility region  $\{\vartheta : \overline{\Pi}_y(\{\vartheta\}) > \alpha\}$  has coverage probability  $\geq 1 - \alpha$ .
- 
- IM validity  $\implies$  usual frequentist validity
  - Connection is mutually beneficial:
    - IMs help with interpretation of frequentist output
    - calibration makes IM's  $(\underline{\Pi}_y, \overline{\Pi}_y)$  real-world relevant

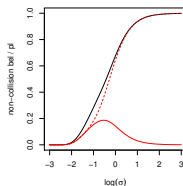
# Construction of valid IMs

- $Y = a(\theta, U)$ ,  $U \sim P_{\text{known}}$ .
- If we could observe  $U$ , then inference on  $\theta$  is trivial.
- Idea: use a suitable random set to “guess” the unobserved  $U$ .
- Push random set through to a data-dependent possibility measure on  $\theta$ -space.
- No time for details :(

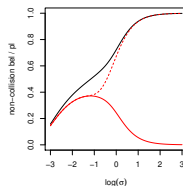


# Satellite collision, revisited

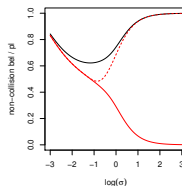
- Satellite collision illustration:  $A = \{\text{non-collision}\}$ 
  - $\Pi_y(A)$
  - $\underline{\Pi}_y(A)$  and  $\overline{\Pi}_y(A)$  (dashed)
- Gap between  $\underline{\Pi}_y(A)$  and  $\overline{\Pi}_y(A)$  is increasing in  $\sigma$



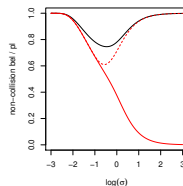
(h) Close



(i) Kinda close



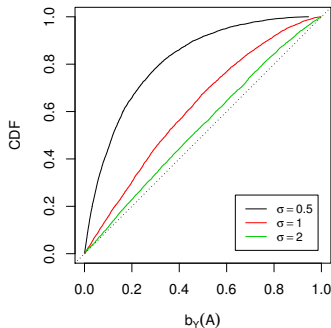
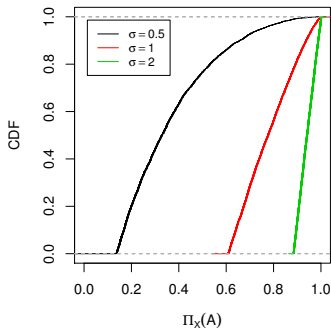
(j) Kinda far



(k) Far

# Satellite collision, revisited

- Same example: collision course,  $A = \{\text{non-collision}\}$ .
  - Left is the CDF of  $\Pi_Y(A)$
  - Right shows the same for  $\underline{\Pi}_Y(A)$
- Validity prevents  $\underline{\Pi}_Y(A)$  from being too large.



- The lower/upper probability perspective is the right way to think about frequentist/NP inference.
- Mathematical facts:<sup>5</sup>
  - p-values measure the *plausibility* of the null hypothesis
  - confidence regions are collections of parameter values that are sufficiently *plausible*
- I teach this “plausibility” interpretation to my students:  
*“Your discussion of plausibility in ST503 made it easier for me to explain what a p-value is when I was interviewing. Thank you for the new perspective on the subject!”*

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<sup>5</sup>M. (2021), [researchers.one/articles/21.01.00002](https://researchers.one/articles/21.01.00002)



- [researchers.one/articles/20.01.00010](https://researchers.one/articles/20.01.00010)

Valid inferential models for prediction in supervised  
learning problems

Leonardo Cella\* and Ryan Martin†

December 7, 2021

- Not quite finished...

Inferential models and the decision-theoretic  
implications of the validity property

Ryan Martin\*

November 19, 2021

# What's next?

- Imprecise prob is flexible, allows for *partial prior info*.
- High-dim problems are my motivation:
  - structural assumptions, e.g., sparsity, parsimony, etc.
  - this is just partial prior information
- How to use imprecise prob to incorporate this in a way that's both valid and efficient relative to prior info?

Towards a theory of valid inferential models with  
partial prior information

Ryan Martin\*

December 15, 2021

- *Foundations matter!* Inference can be unreliable when prior ignorance is converted to a precise probability
- Reliability/validity is crucial to the logic of inference
- False confidence theorem: validity requires imprecise prob
- “Consonant”<sup>6</sup> IMs seem especially important for theory, interpretation, and computation
- Goal is an entire framework for probabilistic inference, prediction, and decision-making:
  - validity guarantees
  - incorporate (partial) prior info when available.
- New and exciting territory.

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<sup>6</sup>Based on nested random sets

*Thanks for your attention!*

`www4.stat.ncsu.edu/~rgmarti3`

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